IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Nguyen Xuan Luu
09/May/2026

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
  - Collect the data
  - Data Wrangling
  - Explore Data Analysis (EDA)
  - Visual Analytics
  - Predictive Analysis
- Summary of all results
  - We can collect data from public source in the internet
  - EDA and Visual Analytics help us understand the patterns of data
  - Predictive Analysis is important for applying patterns to predict the future results

Introduction

- Project background and context

In this capstone project, we aim to predict the successful landing of the Falcon 9 first stage. SpaceX advertises rocket launches at a significantly lower cost compared to other providers, largely due to their ability to reuse the first stage of the rocket. By accurately predicting landing success, we can estimate launch costs and provide valuable insights for companies bidding against SpaceX.

- Problems you want to find answers
  - What factors influence the successful landing of the Falcon 9 first stage?
  - How can we accurately predict the landing outcome using machine learning models?
  - Which machine learning model performs best in predicting the landing success?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:

Data from Space X was obtained from 2 sources:

  - Space X API (https://api.spacexdata.com/v4/rockets/)
  - Web Scraping

- Perform data wrangling

Collected data was enriched by creating a landing outcome label based on outcome data after summarizing and analyzing features

- Perform exploratory data analysis (EDA) using visualization and SQL

- Perform interactive visual analytics using Folium and Plotly Dash

Methodology

Executive Summary

- Perform predictive analysis using classification models

Data that was collected until this step were normalized, divided in training and test data sets and evaluated by four different classification models, being the accuracy of each model evaluated using different combinations of parameters.

Data Collection

- Describe how data sets were collected.
- Data sets were collected from Space X API (https://api.spacexdata.com/v4/rockets/) and from Wikipedia (https://en.wikipedia.org/wiki/List_of_Falcon/_9/_and_Falcon_Heavy_launches), using web scraping technics.

Data Collection – SpaceX API

- SpaceX offers a public API from where data can be obtained and then used;
- This API was used according to the flowchart beside and then data is persisted.
- GitHub URL:

Data Collection - Scraping

- Data from SpaceX launches will be obtained from Wikipedia
- Use pandas and beautiful soup to download, extract and format data to pandas data frame
- GitHub URL

Data Wrangling

- Exploratory Data Analysis (EDA) was performed on the dataset.
- Summarize launches per site, occurrences of each orbit and occurrences of mission outcome per orbit type were calculated.
- Creating the landing outcome label from Outcome column.
- GitHub URL:

EDA with Data Visualization

- To explore data, scatterplots and barplots were used to visualize the relationship between pair of features: Payload Mass versus Flight Number, Launch Site versus Flight Number, Launch Site versus Payload Mass.
- GitHub URL

EDA with SQL

- The following SQL queries were performed:
    - Names of the unique launch sites in the space mission;
    - Top 5 launch sites whose name begin with the string 'CCA';
    - Total payload mass carried by boosters launched by NASA (CRS);
    - Average payload mass carried by booster version F9 v1.1;
    - Date when the first successful landing outcome in ground pad was achieved;
    - Names of the boosters which have success in drone ship and have payload mass between
    - 4000 and 6000 kg;
    - Total number of successful and failure mission outcomes;
    - Names of the booster versions which have carried the maximum payload mass;
    - Failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015; and Rank of the count of landing outcomes (such as Failure (drone ship) or Success (groundpad)) between the date 2010-06-04 and 2017-03-20.
- GitHub URL:

Build an Interactive Map with Folium

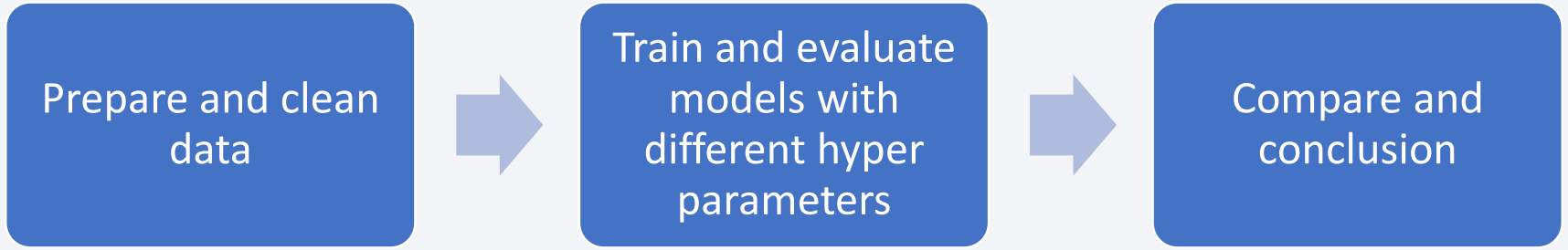
- Markers, circles, lines and marker clusters were used with Folium Maps
  - Markers indicate points like launch sites;
  - Circles indicate highlighted areas around specific coordinates, like NASA Johnson Space Center;
  - Marker clusters indicates groups of events in each coordinate, like launches in a launch site; and Lines are used to indicate distances between two coordinates.
- GitHub URL:

Build a Dashboard with Plotly Dash

- The following graphs and plots were used to visualize data
  - Percentage of launches by site
  - Payload range
  - This combination allowed to quickly analyze the relation between payloads and launch sites, helping to identify where is best place to launch according to payloads.
- GitHub URL:

Predictive Analysis (Classification)

- Four classification models were compared: logistic regression, support vector machine, decision tree and k nearest neighbors.
- Github URL:

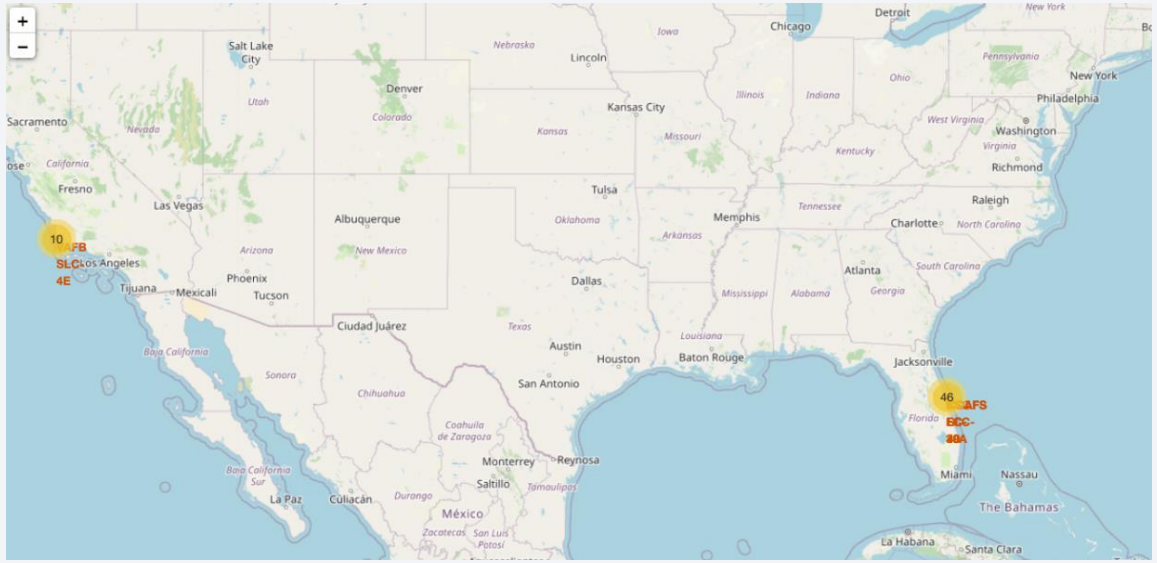


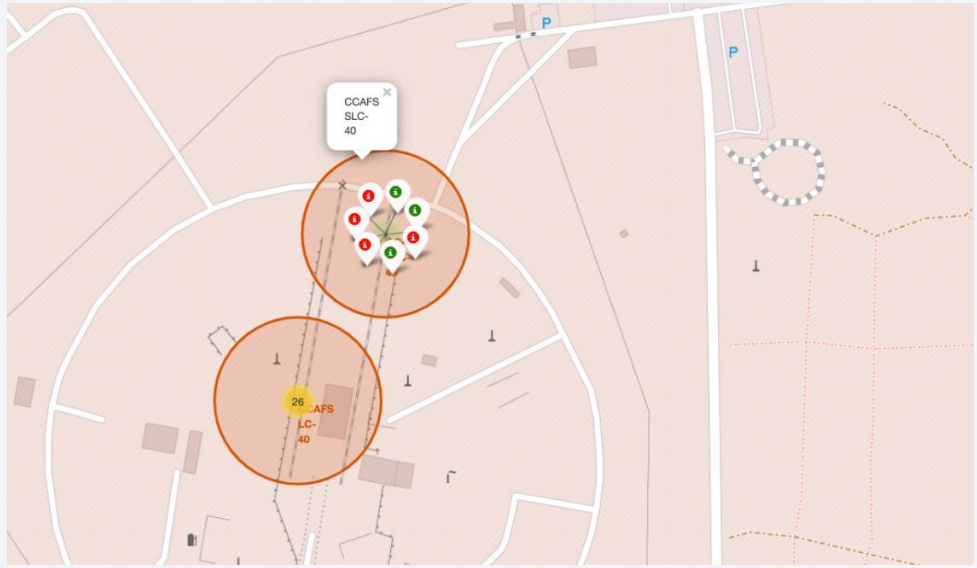
Results

- Exploratory data analysis results
  - There are 4 different launch sites
  - The first success landing outcome happened in 2015
  - The average payload of F9 v1.1 booster is 2,928 kg
  - Many Falcon 9 booster versions were successful at landing in drone ships having payload above the average
  - The number of landing outcomes became as better as years passed

Results

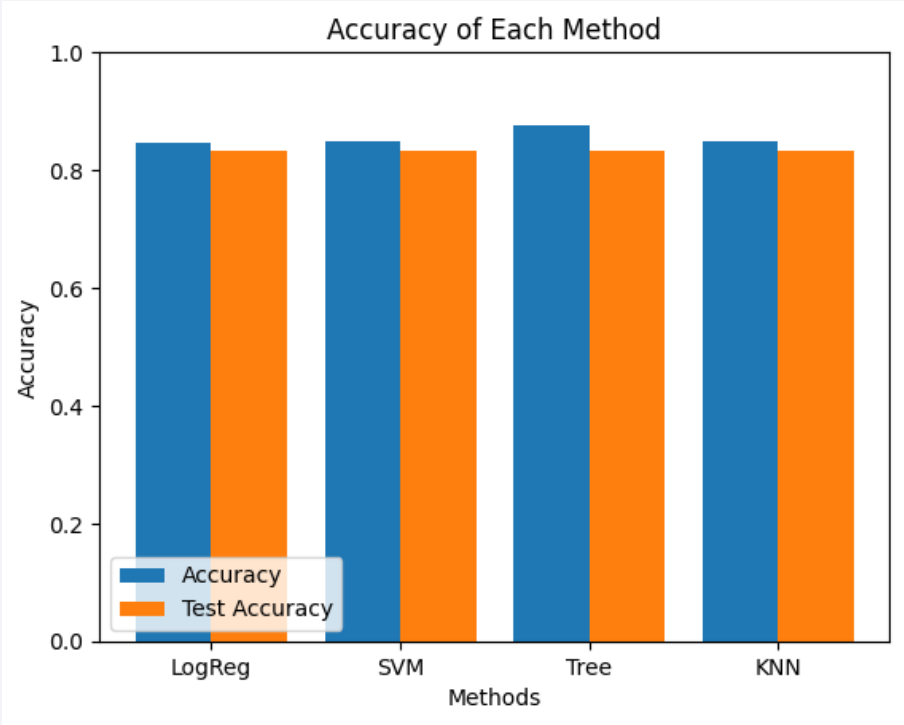
- Interactive analytics demo in screenshots





Results

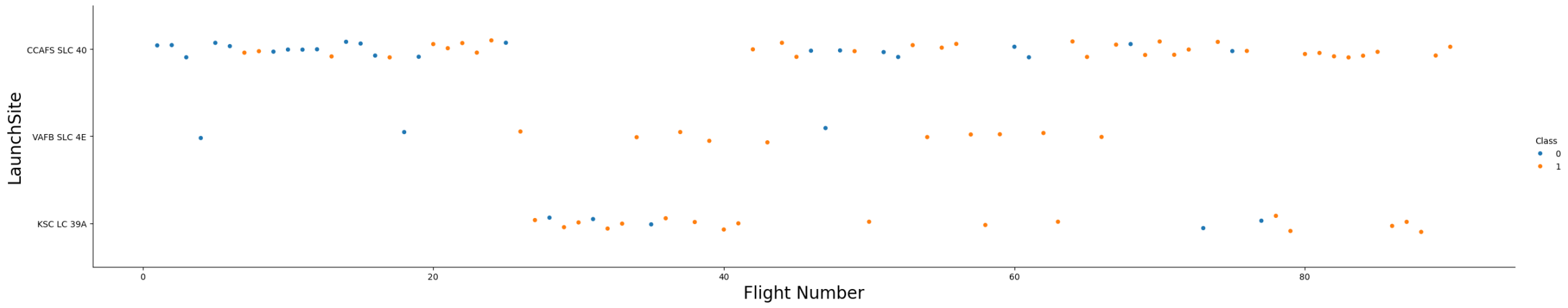
- Predictive analysis results
  - Decision Tree Classifier is the best model to predict successful landings, having accuracy over 87%



Section 2

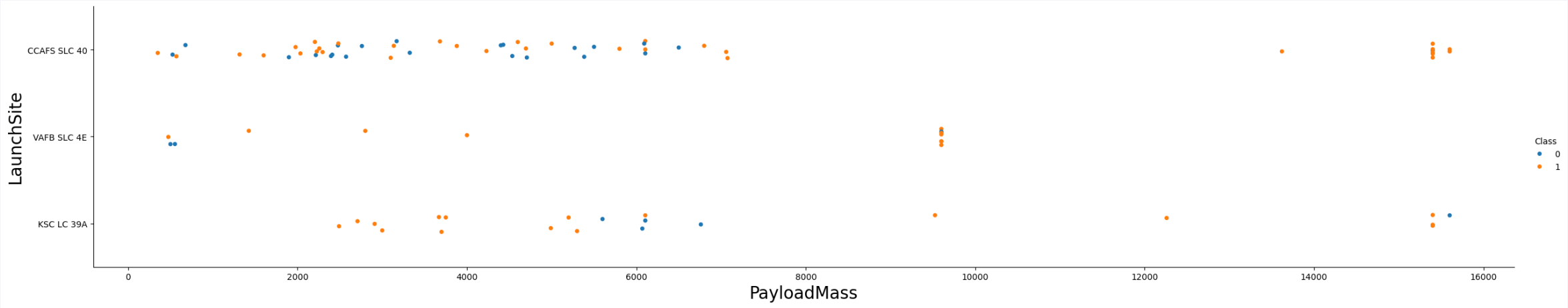
Insights drawn from EDA

Flight Number vs. Launch Site



- The best launch site is CCAF5 SLC 40
- The second is VAFB SLC 4E
- The third is KSC LC 39A
- All the launch sites are improved over time

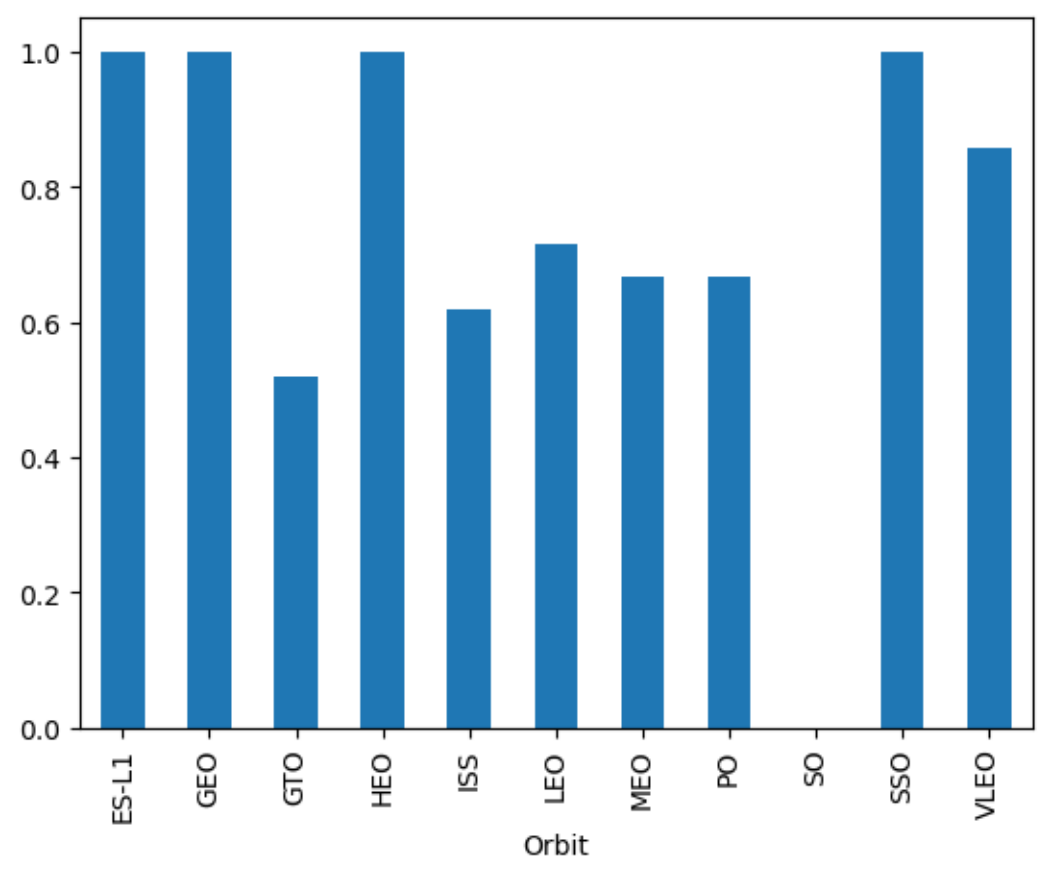
Payload vs. Launch Site



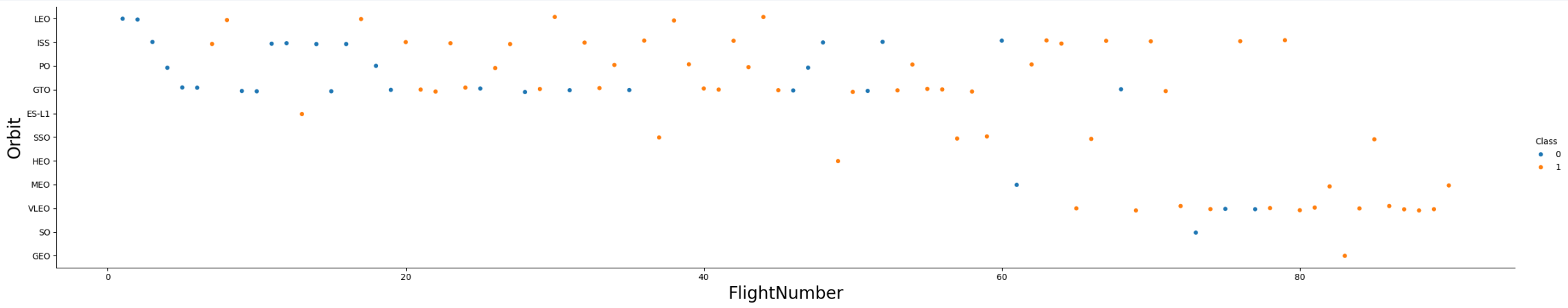
- The launch site CCAF5 SLC 40 has good success rate with over 8000kg payload
- The launch site VAFB SLC 4E has few results with below 6000kg payload
- The launch site KSC LC 39A has good success rate with below 4000kg payload

Success Rate vs. Orbit Type

- ES-L1, GEO, HEO and SSO has 100% success rate
- GTO has lowest success rate

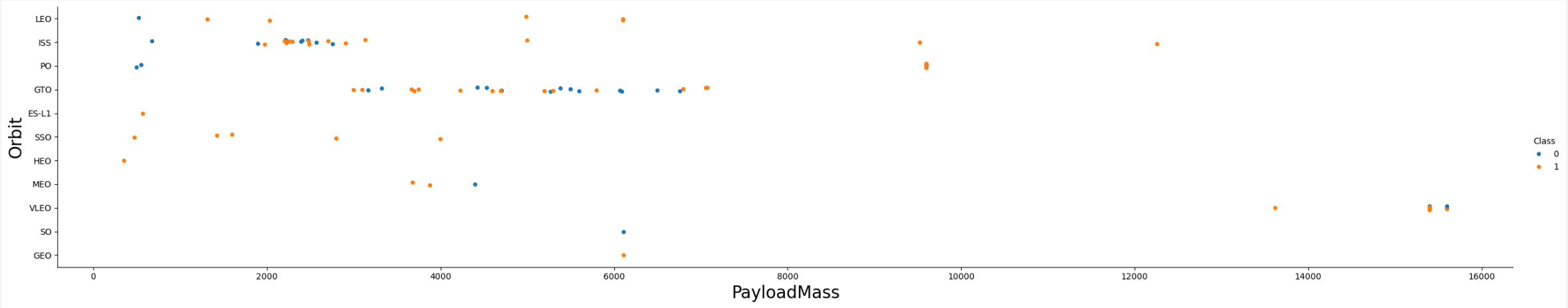


Flight Number vs. Orbit Type



- The VLEO has much focus in the recently time
- The LEP, ISS, PO and GTO has many flight numbers overall

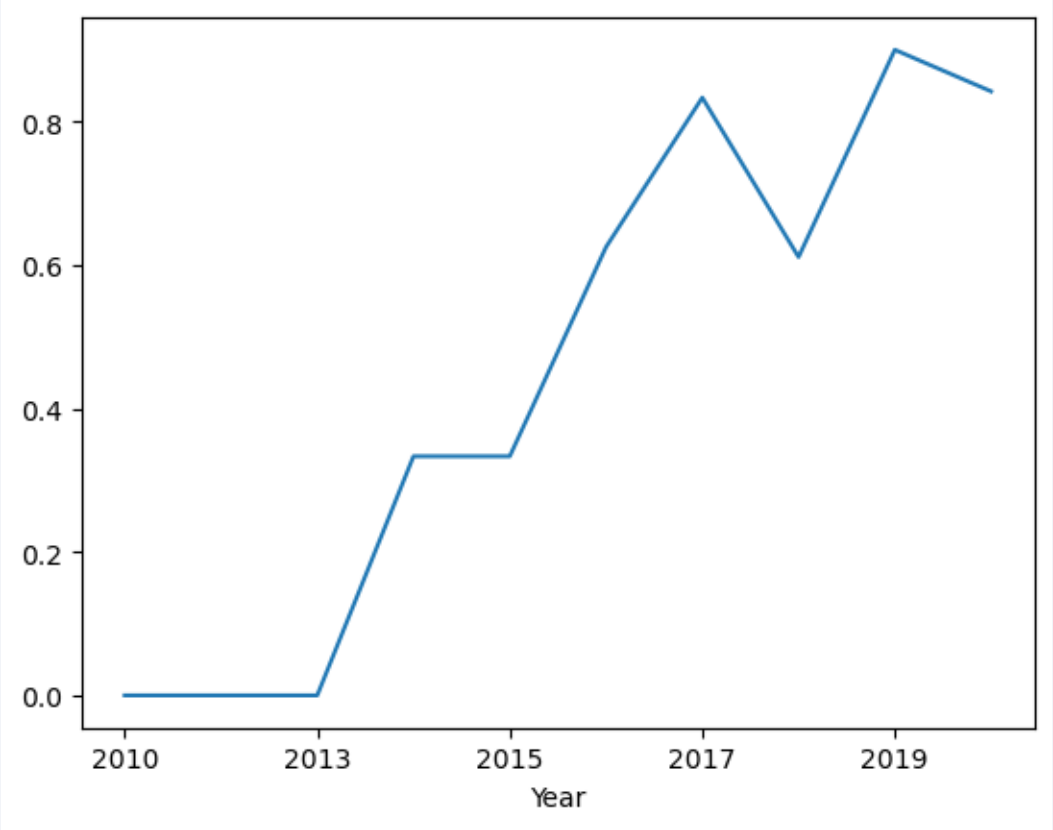
Payload vs. Orbit Type



- There are so many results for ISS at 2000 kg payload
- VLEO has results at over 14000kg payload

Launch Success Yearly Trend

- Success rate are increased from 2013 to 2020
- The best success rate is at 2019



All Launch Site Names

- The names of the unique launch sites
  - CCAFS LC-40
  - CCAFS SLC-40
  - KSC LC-39A
  - VAFB SLC-4E
- The query select distince launch_site from table data

```
%sql SELECT DISTINCT LAUNCH_SITE FROM SPACEXTBL ORDER BY 1;
```

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- The query select with like clause and limit 5 record

```
%sql SELECT * FROM SPACEXTBL WHERE LAUNCH_SITE LIKE 'CCA%' LIMIT 5;
```

Total Payload Mass

- Calculate the total payload carried by boosters from NASA

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) AS TOTAL_PAYLOAD FROM SPACEXTBL WHERE PAYLOAD  LIKE '%CRS%';
```

```
 * sqlite:///my_data1.db
Done.
```

TOTAL_PAYLOAD
111268

- The query use sum function to calculate the total payload

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1

In [13]:
```
%sql SELECT AVG(PAYLOAD_MASS__KG_) AS AVG_PAYLOAD FROM SPACEXTBL WHERE BOOSTER_VERSION = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

Out[13]:

AVG_PAYLOAD
2928.4

- The query use AVG to calculate the mean of payload mass

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad

```
[15]: %sql SELECT MIN(DATE) AS FIRST_SUCCESS_GP FROM SPACEXTBL WHERE LANDING_OUTCOME = 'Success (ground pad)';

 * sqlite:///my_data1.db
Done.
```

[15]:

FIRST_SUCCESS_GP
2015-12-22

- The query use min function to find the first successful landing outcome

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- The query show the distince value of bosster_version to show the lists

```
%sql SELECT DISTINCT BOOSTER_VERSION FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ BETWEEN 4000
AND 6000 AND LANDING_OUTCOME = 'Success (drone ship)';
```

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes

```
%sql SELECT MISSION_OUTCOME, COUNT(*) AS QTY FROM SPACEXTBL GROUP BY MISSION_OUTCOME ORDER BY MISSION_OUTCOME;
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	QTY
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- The query use group by clause and count function to calculate the total number

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass

Booster_Version
F9 B5 B1048.4
F9 B5 B1048.5
F9 B5 B1049.4
F9 B5 B1049.5
F9 B5 B1049.7
F9 B5 B1051.3
F9 B5 B1051.4
F9 B5 B1051.6
F9 B5 B1056.4
F9 B5 B1058.3
F9 B5 B1060.2
F9 B5 B1060.3

- The query uses distinct to find the list

```
%sql SELECT DISTINCT BOOSTER_VERSION FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT
MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL) ORDER BY BOOSTER_VERSION;
```

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

Month	Booster_Version	Launch_Site
01	F9 v1.1 B1012	CCAFS LC-40
04	F9 v1.1 B1015	CCAFS LC-40

- The query uses group by clause with conditions to find the list

```
%sql SELECT substr(Date, 6,2) as Month,BOOSTER_VERSION, LAUNCH_SITE FROM SPACEXTBL WHERE
LANDING_OUTCOME = 'Failure (drone ship)' AND substr(Date,0,5)='2015' group by substr(Date, 6,2)
order by 1;
```

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Landing_Outcome	QTY
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

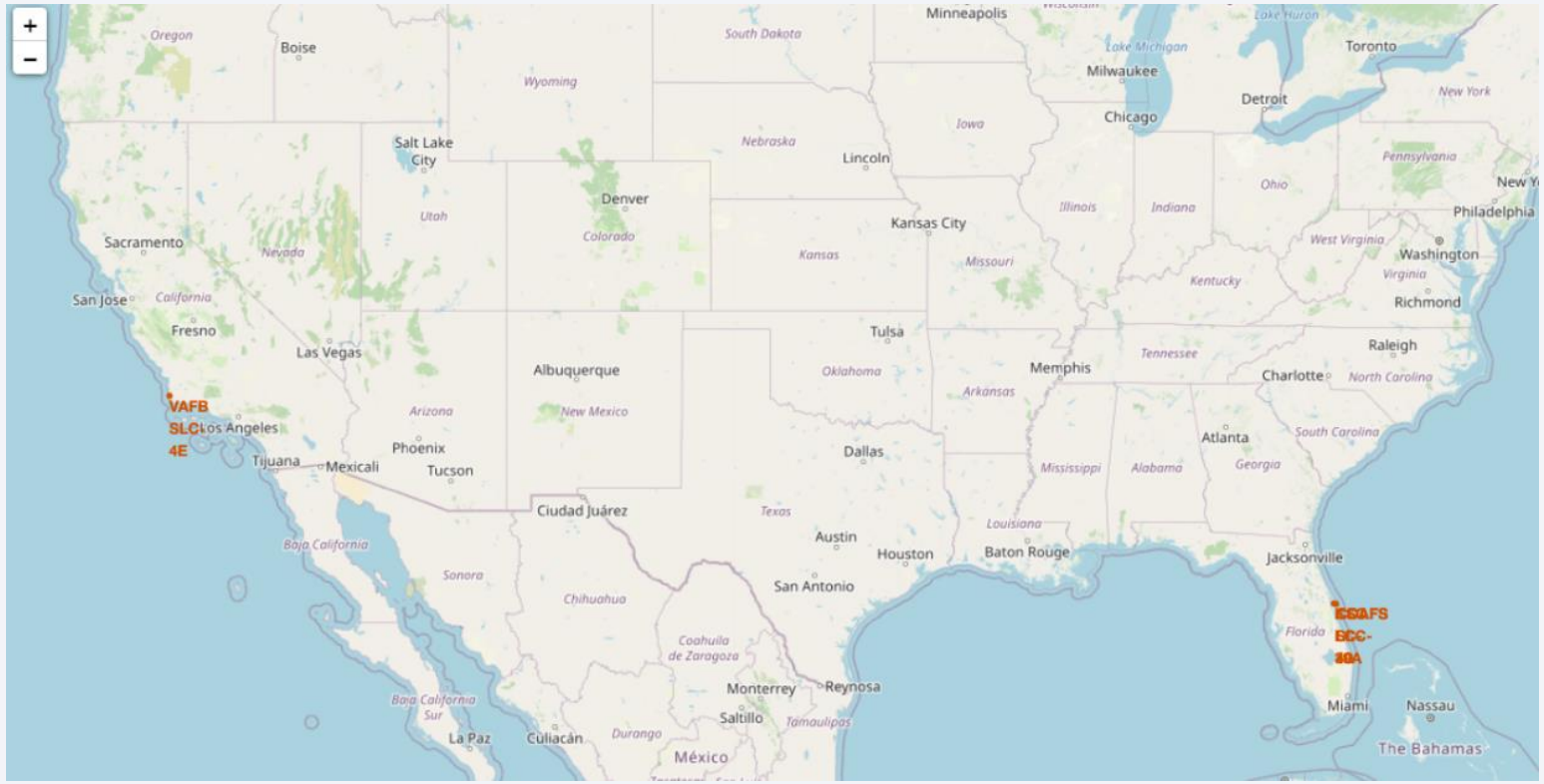
- The query use group by clause and order by .. desc to get the rank table

```
%sql SELECT LANDING_OUTCOME, COUNT(*) AS QTY FROM SPACEXTBL WHERE DATE BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY LANDING_OUTCOME ORDER BY QTY DESC;
```

Section 3

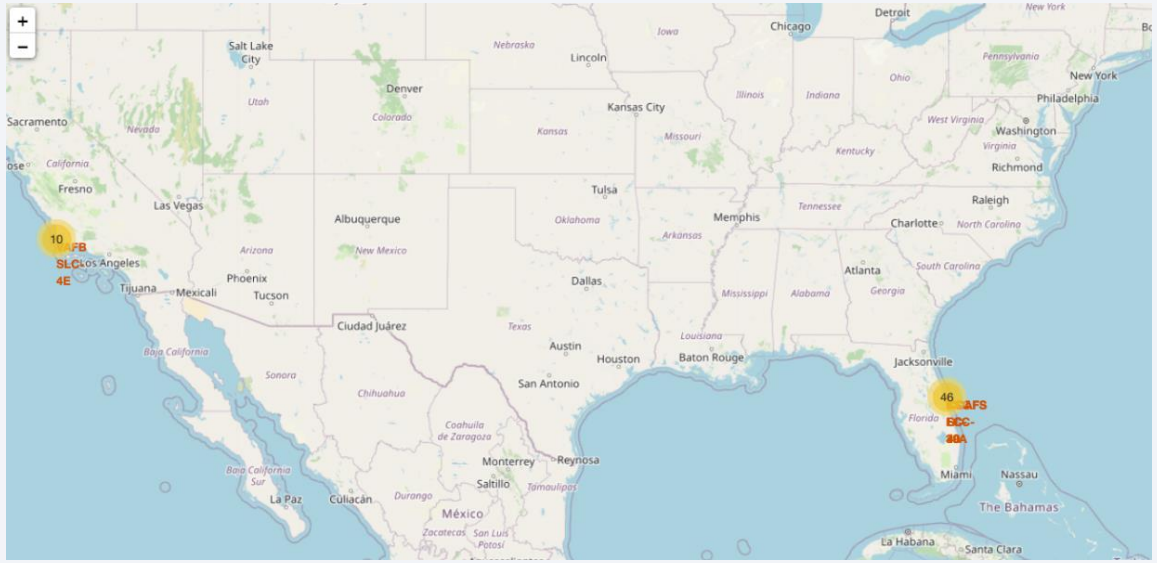
Launch Sites Proximities Analysis

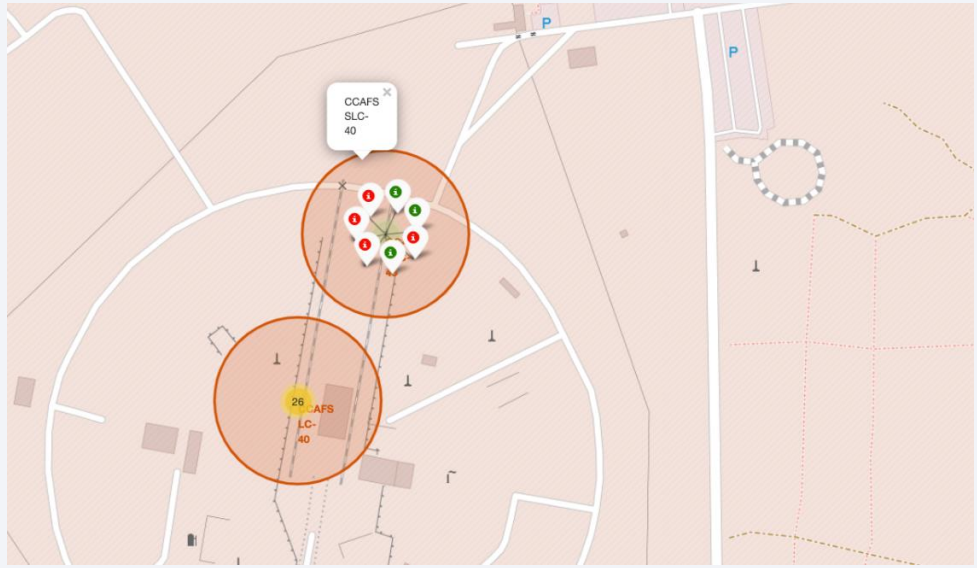
All launch site



- The launch sites are located at near sea and in the middle area

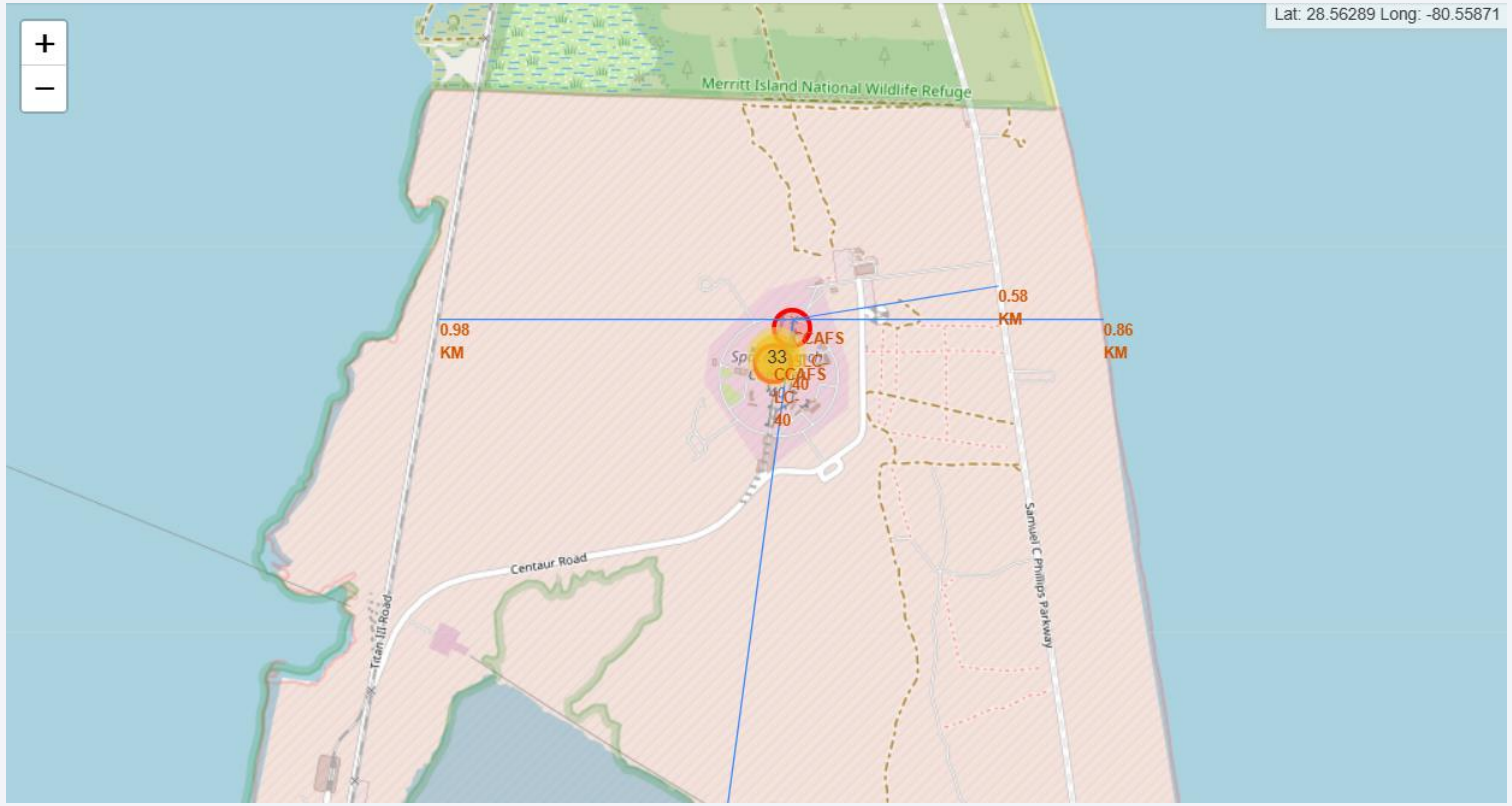
Launch Outcomes by Sites





- The red markers show failure and green markers show successful launch sites

Safety and Distance

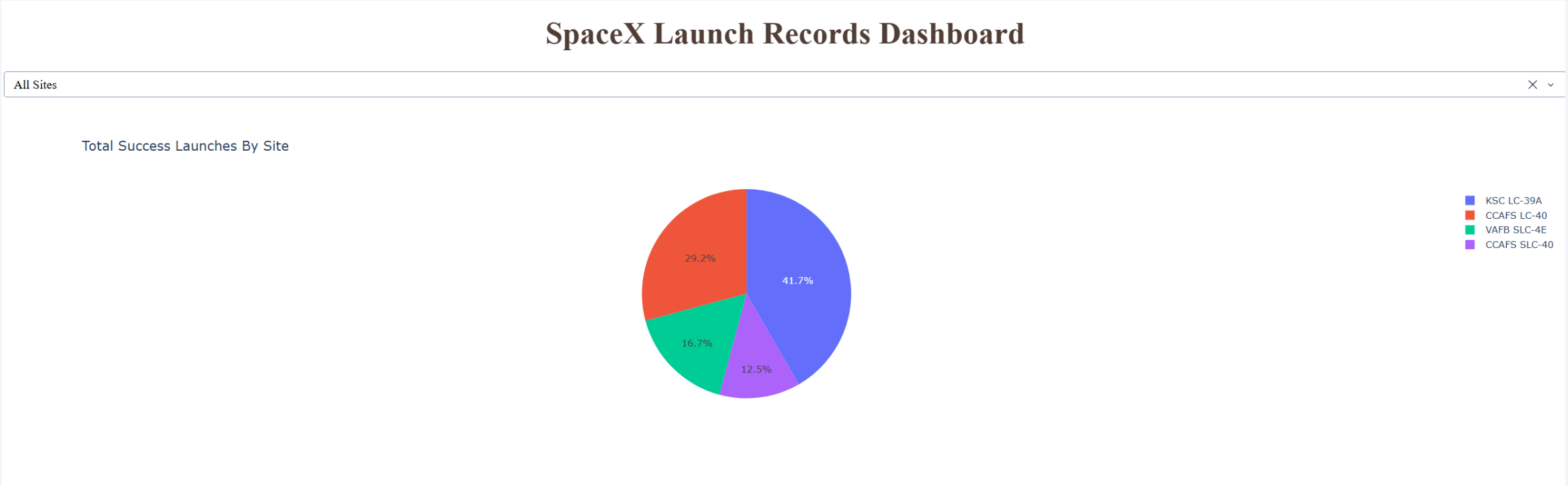


- The launch sites is near highway and railway but far from city

Section 4

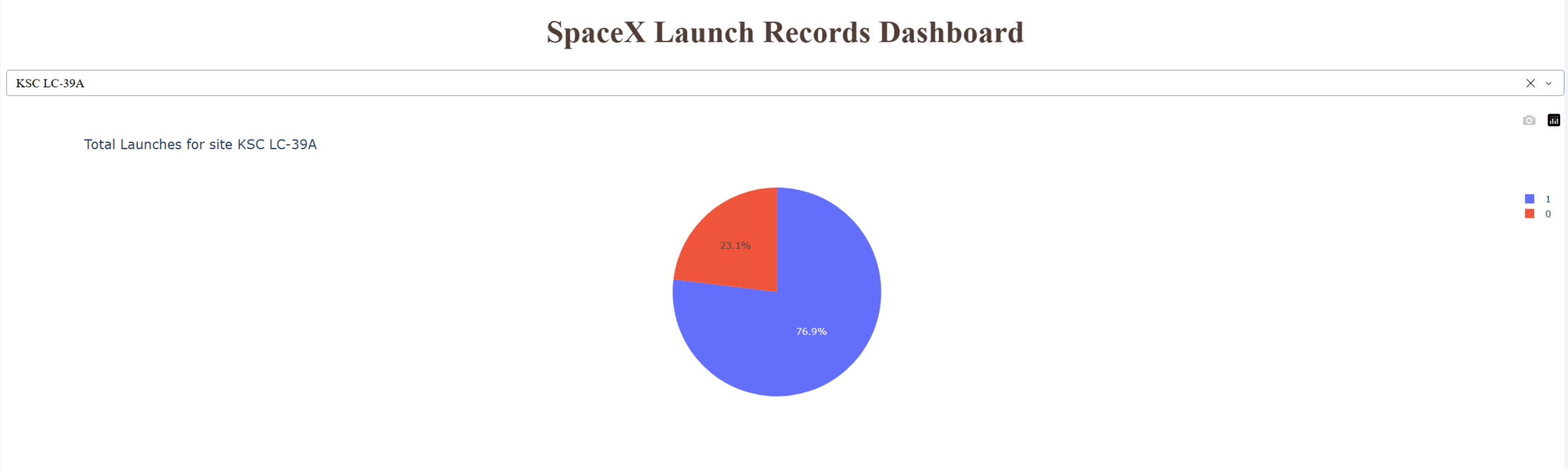
Build a Dashboard with Plotly Dash

Launch success count



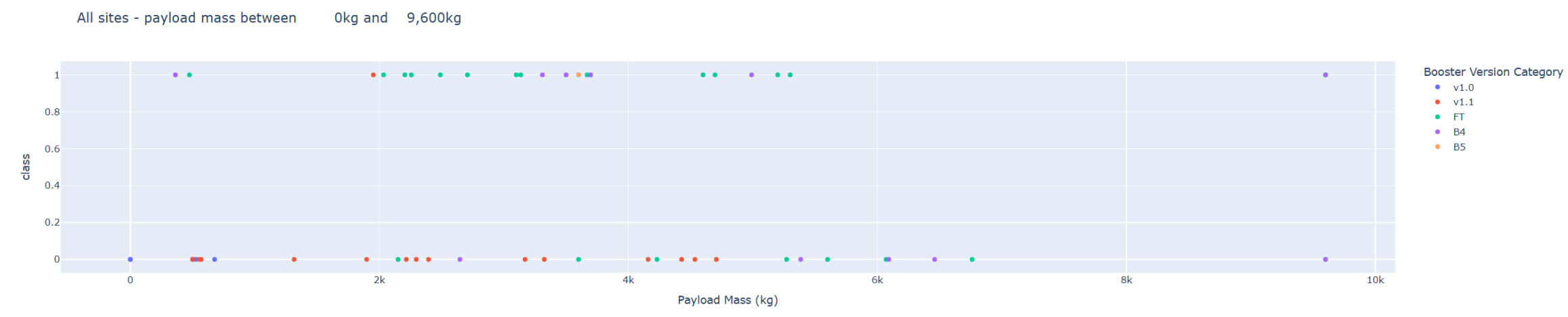
- With visualization results, the site is the import factor for successful rate

Launch site with highest launch success ratio



- The success ratio is 76,9%

Payload vs. Launch Outcount



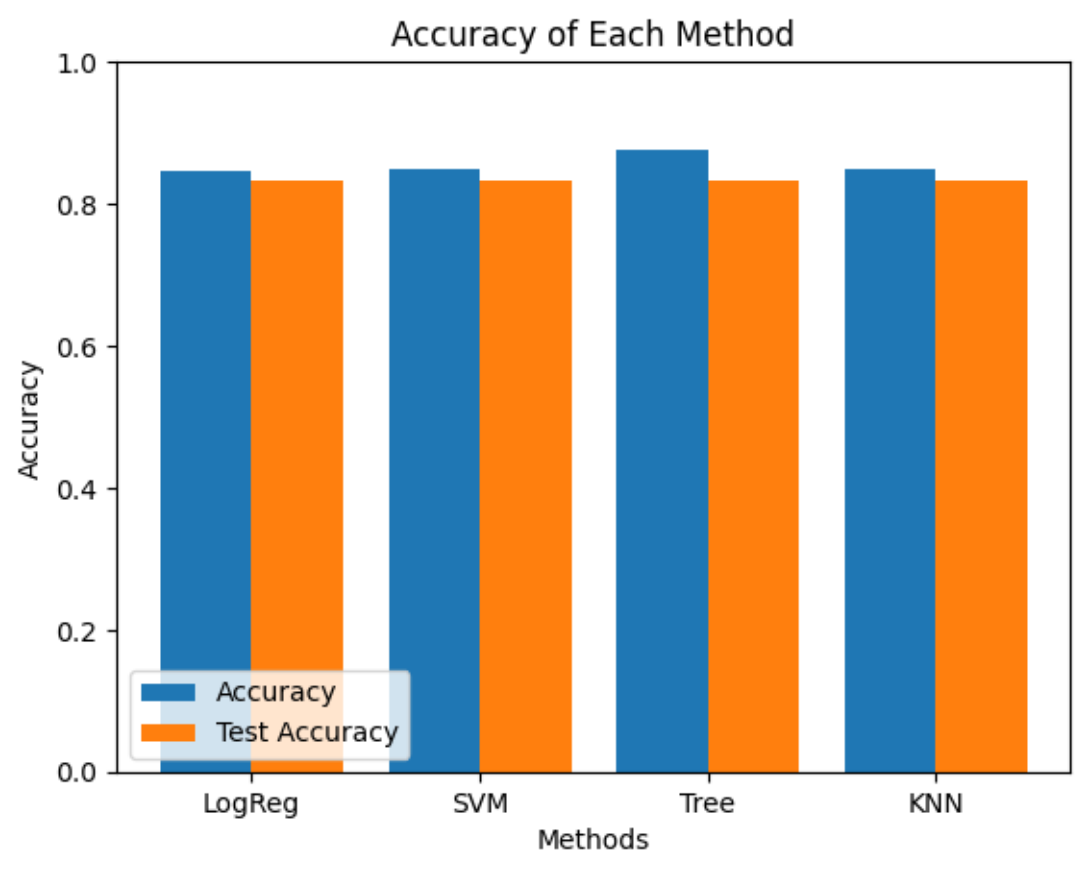
- FT Booster with below 6000kg payload gains the best successful result

Section 5

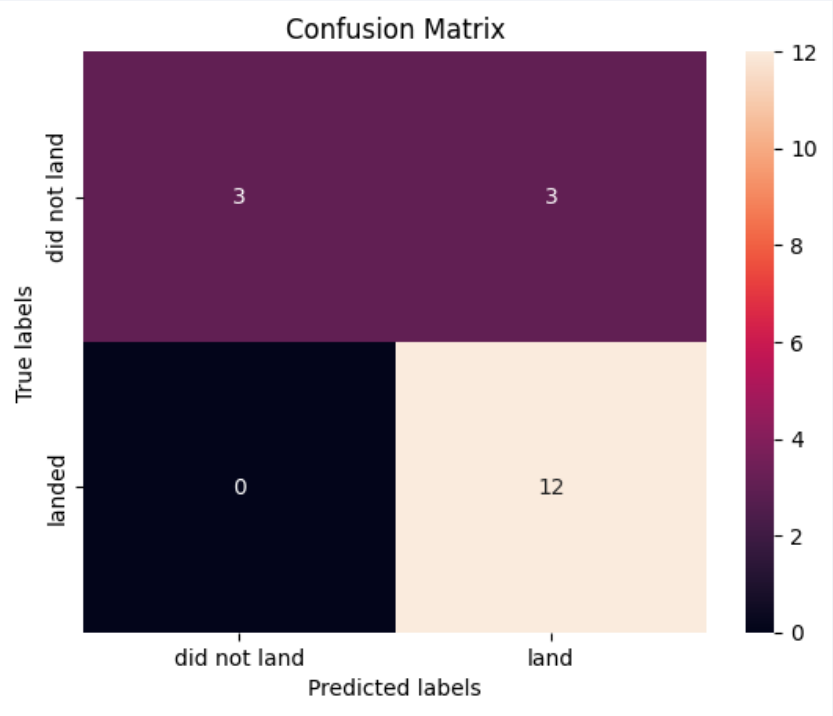
Predictive Analysis (Classification)

Classification Accuracy

- The Decision Tree Classifier has the best accuracy with 87,5%



Confusion Matrix



- Confusion matrix of Decision Tree Classifier has good accuracy with land label and land true results

Conclusions

- Decision Tree Classifier can be used to predict successful landings and increase profits.
- Successful landing outcomes are improved over time.

Appendix

- Github project: https://github.com/xuanluudata/applied-data-science-ibm-coursera/

Thank you!